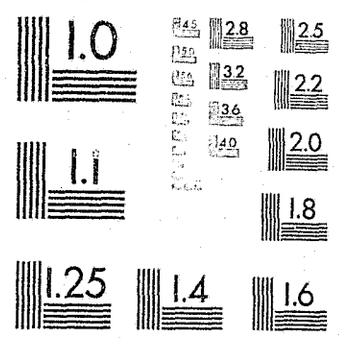
1-A-43407/85



MICROCOPY RESOLUTION TEST CHART
NATIONAL BUREAU OF STANDARDS
STANDARD REFERENCE MATERIAL 1010a
(ANSI and ISO TEST CHART No. 2)

7



Form 1 Regulation 9

COMMONWEALTH OF AUSTRALIA

Patents Act 1952

FEE STAMP TO VALUE OF $ 165 ATTACHED
MAIL OFFICER

AUSTRALIA
7 JUN 1985
PATENT OFFICE

APPLICATION FOR A STANDARD PATENT

We, ICI AMERICAS INC, of New Murphy Road and Concord Pike, Wilmington, Delaware 19897, United States of America, hereby apply for the grant of a standard patent for an invention entitled:

ALKOXYLATED FATTY ALCOHOLS AS AXILLARY BUD INHIBITORS ON TOBACCO PLANTS

which is described in the accompanying complete specification.

Details of basic application(s):
Number of basic application: 06/625,220
Name of Convention country in which basic application was filed: UNITED STATES OF AMERICA
Date of basic application: June 27, 1984

Our address for service is: Patents Section, ICI Australia Limited, P.O. Box 4311, 1 Nicholson Street, Melbourne, Victoria, 3001, Australia.

Dated this 6th day of June 1985

ICI AMERICAS INC.
By its Patent Attorney

D W Berryman

To: The Commissioner of Patents

ATL 1512

FORM 8

Regulation 12(2)

Patents Act 1952

DECLARATION IN SUPPORT OF A CONVENTION APPLICATION FOR A PATENT

In support of the Application made by ATKEMIX INC. , for a patent for an invention entitled:-

ALKOXYLATED FATTY ALCOHOLS AS AXILLARY BUD INHIBITORS ON TOBACCO PLANTS

I, Allen Dickinson Dabney OF ATKEMIX INC. 70 Market Street, Brantford, Ontario, Canada

do solemnly and sincerely declare as follows:-

1. I am authorised by ATKEMIX INC. , the applicant for the patent to make this declaration on its behalf.

2. The basic application as defined by section 141 of the Act was made in the United States of America by Alan Walter NICHOLLS and Edwin Caesar KLINE on the 27thday of June 1984.

U.S. Serial No. 06/625,220 .

3. Alan Walter NICHOLLS and Edwin Caesar KLINE ~~is~~ are the actual inventors of the invention and the facts upon which ATKEMIX INC. , is entitled to make the application are as follows:

The said ATKEMIX INC. is the assignee of the actual inventors in respect of the invention and of the priority right arising from the basic application.

4. The basic application referred to in paragraph 2 of this Declaration was the first applications made in a Convention country in respect of the invention the subject of the application

Declared at Wilmington, Delaware, U.S.A. this 17 day of June 1985

.....

Allen Dickinson Dabney
Vice President

(12) AUSTRALIAN PATENT ABSTRACT
(19) AU (11) AU-A-43407/85

(54) ALKOXYLATED FATTY ALCOHOL BUD INHIBITORS FOR TOBACCO

(71) ATKEMIX INC.

(21) 43407/85 (22) 7.6.85 (24) 27.6.84
(31) 625220 (32) 27.6.84 (33) US
(43) 2.1.86
(51)4 A01N 49/00

(72) ALAN WALTER NICHOLLS AND EDWIN CAESAR KLINE

(57) Claim

1. A method for inhibiting the growth of suckers in flue-cured tobacco plants which comprises applying to the plant shortly before or after topping an effective amount of a liquid alkoxylated saturated C6-C10 alcohol having a hydrophilic-lipophilic balance value of 8-14, wherein said alkoxylated alcohol has the general formula $RO(CH_2CHR^1O)_nH$ wherein R is an alkyl group having 6-10 carbon atoms, R^1 is H, CH_3, C_2H_5 and n is 2-10.

105(84)/32

COMMONWEALTH OF AUSTRALIA

Form 10

PATENTS ACT 1952-1969

COMPLETE SPECIFICATION

(ORIGINAL)

FOR OFFICE USE

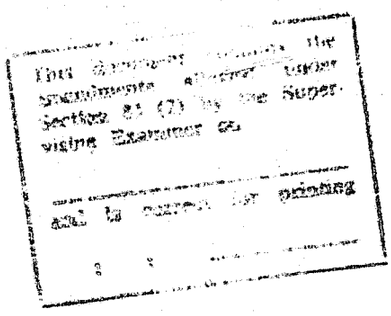
Class Int. Class

Application Number:
Lodged:

Complete Specification—Lodged:
Accepted:
Published:

Priority:

Related Art:



43407/85

TO BE COMPLETED BY APPLICANT

Name of Applicant: ~~ICI AMERICAS INC.~~
ATKEMIX INC

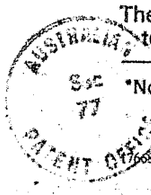
Address of Applicant: New Murphy Road and Concord Pike
Wilmington, Delaware 19897
United States of America

Actual Inventor: Alan Walter NICHOLLS
Edwin Caesar KLINE

Address for Service: Patents Section
ICI Australia Limited
1 Nicholson Street
MELBOURNE 3000 Victoria Australia

Complete Specification for the invention entitled:
ALKOXYLATED FATTY ALCOHOLS AS AXILLARY BUD INHIBITORS ON TOBACCO PLANTS

The following statement is a full description of this invention, including the best method of performing it known to me:—*



*Note: The description is to be typed in double spacing, pica type face, in an area not exceeding 9⅞" in depth and 6½" in width, on tough white paper of good quality and it is to be inserted inside this form.

W. G. Murray, Government Printer, Canberra

ALKOXYLATED FATTY ALCOHOLS AS AXILLARY BUD INHIBITORS ON TOBACCO PLANTS

This invention relates to the control of axillary buds (suckers) in tobacco plants by the application of contact-type chemical sucker control agents. In particular it is directed to the use of alkoxylated fatty alcohols having 6-10 carbon atoms and a hydrophilic/lipophilic balance (HLB) of 8-14.

Tobacco leaf quality and yield are greatly increased by "topping" tobacco plants in the mid-bud to late-bud stage of growth and development. This is achieved by removing the flowers and several adjacent top leaves. As soon as the inflorescence is removed, many processes become active. One of these is the accumulation of nicotine and other alkaloids in the leaves, particularly in the upper 2/3 of the plant. Topping also allows the leaves in the upper part of the plant to continue to grow and consequently increase the total yield.

Decapitation (topping) of the plant enhances the growth of axillary buds or "suckers"; these must be removed continuously or permanently in order to achieve the benefits of topping. Removal of the suckers may be accomplished by hand, which is a very time consuming and laborious operation, or by chemical control.

Two types of chemical agents are available: systemic and contact. In the United States of America, it is common to use a combination of both types of sucker control agents. An application of a contact control agent will be followed by the application of a systemic material such as the sorts of maleic hydrazide, followed by another possible application of contact desuckerant. In Canada, owing to the shorter growing seasc:,

the systemic control agents have been found to be less suitable for use resulting in reduced yields, and consequently are not recommended by agricultural advisers and experts. It is also more desirable to use contact type desuckerants because of the lower associated chemical residue in the cured tobacco compared to the use of systemic control agents. Some countries have adopted a legal limitation on the residue in tobacco.

The most common type of contact sucker control agent is composed of approximately 81-87% 1-decanol, or a mixture of 1-decanol/1-octanol, plus 19-13% emulsifier of which the most commonly used is polysorbate 80 as disclosed in Canadian Patent 1023164. The commercial products used for comparative purposes in the examples of this disclosure possessed such a composition.

Sucker control chemicals are normally applied to the plant shortly before or after topping. The buds are only susceptible to chemical control before they start to grow and elongate. With the commercial fatty alcohol/emulsifier formulation, any buds that are longer than approximately 2.5 cm begin to build-up the cuticle layer and become resistant to chemical damage or "burn".

From the above, the importance can be seen of removing all the actively growing suckers. In Ontario the current preferred means of achieving this control is to apply contact chemical desuckerants formulated from fatty alcohols such as disclosed in the related Canadian Patent 1023164.

An object of one aspect of this invention is to provide a more economical method for inhibiting the growth of axillary buds in tobacco plants. Another aspect of this invention is to provide a method for treating tobacco plants which can inhibit sucker development with minimal undesirable

residue deposits in the leaf tissue. Still another aspect of this invention, is to provide compositions which may be applied as convenient aqueous dispersions because they are either self-emulsifying or can be dispersed in water by the addition of a minor portion of phytobland emulsifier.

In general, benefits of this invention can be obtained by applying to tobacco plants immediately before or shortly after topping an alkoxylated saturated C6-C10 alcohol having an hydrophilic-lipophilic balance of 8-14. The term hydrophilic-lipophilic balance (HLB) is commonly used to characterize alkoxylated materials. The value may be determined empirically by comparison to surfactants of known HLB values by emulsifying materials with a known HLB requirement; alternatively, the HLB value of a surfactant may be calculated. For ethoxylated compounds, the HLB value may be calculated by dividing the percent ethylene oxide in the molecule by a factor of 5. The HLB values listed for the examples in Table 1 were calculated for ethoxylated alcohols by the above mentioned method, and determined empirically to a reasonable proximation for propoxylated adducts. Additional methods for determining HLB are set forth in "The Atlas HLB System" 4th Printing (1963) ICI Americas Inc., Wilmington, Delaware USA and in Petrawski, G. E. and Vanotta, J. R., J. Am. Chem. Soc. pages 284-289 August 1973.

Although various alkoxylated alcohols of the type disclosed have been found to be effective tobacco sucker control agents, best results, and ones free from undesirable side effects which could result in detrimental economic value to the tobacco quality or yield, were achieved with a polyoxyethylene 1-octanol which has an average oxyethylene chain

length of approximately 3; this compound is thus considered to be a preferred agent of the type disclosed in this invention.

In general the agents used in the invention are produced using well known techniques by reacting ethylene or propylene oxide with saturated alcohols having 6-10 carbon atoms such as 1, 2 or 3-hexanol, 1, 2, 3 or 4 octanol, 1, 2 or 3 decanol and mixtures and isomers thereof in a pressurized reactor to produce liquid compounds having the general formula $RO(CH_2\text{-}CH_2R^1O)_nH$ where R is an alkyl group having 6-10 carbon atoms, R^1 is H or $-CH_3$, $-C_2H_5$ and n is 2-10. The compound may also contain mixtures of ethylene oxide and propylene oxide when the alcohol is reacted with mixtures of the oxides or may be made reacting the alcohols in sequence with the oxides. For example when 1 mol of 1-octanol is reacted with 1 mol of ethylene oxide and 2 mols of propylene oxide the composition is written polyoxyethylene (1) polyoxy-propylene (2) -1 octanol, or poe (1) pop (2) -1 octanol having the formula $C_8H_{17}O(C_2H_4O)(C_2H_3(CH_3)O)_2H$.

The agents of the invention are usually free flowing liquids which are clear to slightly cloudy. If they are made in the presence of an alkaline catalyst the liquid may be too alkaline for use and therefore must be neutralized with acid to a pH of about 5-7 although in some instances a slightly higher or lower pH may not be phytotoxic. The alcohols used to synthesize the disclosed compositions originate from petroleum fractions; however, it is contemplated that C6-C10 alcohols derived from fats and oils would not materially differ in effectiveness as starting materials for the alkoxylated products.

In some instances it is preferred to apply the desuckering agents as aqueous dispersions employing 5-15% by

weight of an effective phytobland emulsifier or surface active agents.

Among the surface active agents found to be useful in the invention are those classified as anionic, nonionic and cationic surfactants. Specific anionics surfactants which may be useful include the sulfated or sulfonated ethers of long and short chain aliphatic groups such as those having about 15-20 carbon atoms such as the sodium salts of sulfonated alkyl esters, sulfonated glycol esters and sulfonated alkyl substituted amides, alkylated aryl sulfonates and dodecyl-benzene sodium sulfonate), hydroaromatic sulfonates (tetra-hydronaphthalene sodium sulfonate), and soap such as sodium laurate, ammonium stearate and diethanol-ammonium oleate. Useful nonionic surface active agents include monoethers of polyglycol with long chain fatty alcohol such as the reaction products of ethylene oxide or polyethylene glycol with long chain fatty alcohol, monoesters of polyglycols with long chain fatty acids, including reaction products of ethylene oxide or polyethylene glycol with long chain fatty acids, partial esters of polyhydric alcohols with long chain monocarboxylic acids (glycerol monostearate, sorbitan tricleate) and partial and complete esters of long chain monocarboxylic fatty acids with polyglycol ethers or polyhydric alcohols (tristearate acid ester of polyglycol ether of sorbitan). Cationic surfactants useful in the invention include quartenary ammonia salts in which one of the groups attached to the nitrogen has an aliphatic group having at least 8 carbon atoms (trimethylacetyl ammonium halide).

The claims defining the invention are as follows:

Method of Preparation for Polyoxyethylene (3) 1-octanol:

13.4 kg of 1-octanol were loaded into a 36 litre autoclave 40 g sodium hydroxide was added. Air was removed under vacuum and replaced with nitrogen gas. Temperature was raised to 140°C and 13.8 kg ethylene oxide added slowly whilst maintaining a gauge pressure of 206-345 KPa until all the ethylene oxide had been introduced. The reaction was allowed to proceed down to 0 KPa gauge pressure. Full vacuum was pulled at 120°C for 15 minutes to remove any unreacted ethylene oxide. The reaction product was removed from the autoclave, neutralized with with 105 g 85% phosphoric acid, filtered in the presence of a silacious earth filter aid. The product analysis is as follows:

	Analysis
Hydroxyl Number	209
Acid number	0.1
Water content	0.2%
pH (10%)	7.4
Colour Gardner	<1

The following examples are illustrative of aspects of this invention. Data have been taken from both greenhouse and field trials. It will be readily apparent from the compositions which have been found to be effective that substantially any alcohol having a straight or branched alcohol chain length within the range C6-C10, or a blend having a predominance within the said range, will inhibit sucker growth providing that it is alkoxylated so that it possesses an HLB value of 8-14. Most alkoxylated alcohols will render the desired degree of axillary bud growth control, however some are sufficiently phytotoxic at the level required to make the compound unsuitable for commercial acceptance and preference.

7. A method of Claim 1 wherein said

The compositions listed in Table 1 serve to demonstrate but not limit the alkoxylated alcohols which can be used in the practice of the invention.

Table 1

Alkoxylated Alcohols

Composition Number	Av. Composition	HLB
1	poe (8.0) tri-methyl heptanol	13.5
2	poe (5.5) " "	12.0
3	poe (4.0) " "	10.5
4	poe (3.3)	9.5
5	poe (4.0) 1-decanol	10.5
6	50% by wt. pop (4.0) tri-methyl heptanol/50% #1	9.5
7	pop (1.0) poe (4.0) trimethyl heptanol	ca. 10
8	poe (2.5) tri-methyl heptanol	8.2
8A	80% #8 + 20% polysorbate 80	9.5
9	poe (2.8) 2-ethyl hexanol	9.7
10	poe (2.2) 1-hexanol	9.7
11	poe (2.8) 2-octanol	9.7
12	poe (3.0) 1-octanol	10.1
13	50% #2 + 50% #8	9.4

N.B. poe = polyoxyethylene
pop = polyoxypropylene

Unless otherwise stated in the specific examples, tobacco treatments were conducted using the following procedures and conditions:

a) Greenhouse Trials

Plants of tobacco were propagated in flats of muck in a heated greenhouse and when of suitable size the seedlings were transplanted into 10" diameter plastic pots

containing a soil/muck mixture in the ratio of 3:1 v/v. The plants were selected for uniformity prior to treatment with sucker control chemicals at the early bud stage. A measured volume of liquid alkoxylated alcohol, control, or comparative agent was dispersed in 450 liters of water and applied with a modified sprayer and boom arrangement similar to that used for field applications for contact type sucker control chemicals. Approximately a 30 ml dose of spray mixture was applied per plant. Three plants were sprayed per treatment. All observations were compared against a control solution of polysorbate 80 in water, as well as a registered contact type sucker control material. Visual observations for leaf phytotoxicity were based on a predetermined scale of 0 to 10 with 10 being equivalent to no change. Data was collected for sucker numbers and weights on a dry weight basis.

b) Field Trials

A split plot experimental design with four replications was used. Each plot was comprised of two 24.4 meter rows, 61 cm between plants, 107 cm between rows with only one row harvested. Plants were topped immediately prior to application of the materials unless climatic conditions interfered; under such circumstances spraying commenced at the earliest opportunity.

Spray materials were applied by commercial high clearance spray equipment when plants were at the mid-bud stage for development. All treatments were applied at a spray volume of 450 liters/hectare (L/ha) at 100 KPa and a ground speed of 6 kpg. The alkoxylated alcohols are applied at dose rates of .5-6 liters per hectare. Phytotoxicity ratings on a scale of 1 to 10 were taken 10-24 days after spraying, with sucker counts

being taken a few days later. A representative sample of suckers from each replica was used to determine dry weights by freeze drying.

c) Analytical

Reducing sugars were determined by ferricyanide reduction. Total alkaloids were determined colorimetrically by the development of an orange colour in presence of cyanogen bromide.

Tobacco quality is directly associated with accumulative effect of a number of chemical, physical and agronomic parameters. Table 2 below lists the parameters used to assess tobacco quality as reported in the illustrative examples.

Table 2

Tobacco Assessment Parameters

I.	Average number of suckers per plant
II.	Dry weight of suckers per plant
III.	Phytotoxicity rating (0-10)
IV.	Weighted average reducing sugars content (%)
V.	Weighted average total alkaloids content (%)
VI.	Weighted average grade index ($ Can./kg.)
VII.	Yield of tobacco (kg./ha.)
VIII.	Return Index ($ Can./ha.)

Example 1

Composition numbers 1, 2 and 3 were applied to Delhi 76 tobacco plants in August at rates of 5.5, 4.4, 3.3, 2.2 and

1.1 L/ha. Effective sucker control was evident from all three materials down to the 1.1 L/ha level. At a concentration of 5.5 L/ha. severe burns to leaves and axillary buds was observed whereas at 2.2 and 1.1 L/ha. only slight damage to leaf tissue occurred.

Example 2

Composition numbers 3, 4, 5, 6 and 7 as well as commercial contact type sucker control material A were applied to Delhi 34 tobacco plants in a greenhouse at Delhi, Ontario in June. Sucker control efficacy was evaluated after three days and results tabulated below in Table 3.

Table 3

Composition Number	Sucker Control Efficacy						
		Rate (L/ha)					
	Test	0.55	1.1	3.5	5.5	8.0	16.8
3	I	0.0	1.1	4.3	4.8	4.9	-
4	I	0.6	3.2	4.9	4.8	4.8	-
5	I	0.2	0.8	4.2	4.6	4.9	-
6	I	0.1	0.2	1.6	3.2	4.7	-
7	I	0.7	0.6	2.7	4.3	4.9	-
Commercial A	I	-	-	-	-	-	4.8

Score: visual assessment of 0-5 with 5.0 being top score.

Example 3

Composition numbers 3, 4 and 8A, as well as commercial contact sucker control material A and a Control

consisting of a solution of polysorbate 80 in 450 liters of water at a concentration equivalent to that contained in 16.8 liters commercial desuckerant A, were applied to Delhi 34 tobacco plants in a greenhouse in Delhi, Ontario in July. Assessment was made of sucker count, sucker dry weight and leaf phytotoxicity. The results have been tabulated in Table 4.

Table 4

Composition Number	Test Par.	Sucker Control Efficacy Rate (L/ha) 0.55	1.1	3.5	5.5	8.0	16.8
3	I	4.0	1.0	2.0	2.0	0.3	-
	II	7.7	1.9	0.7	1.3	0.1	-
	III	0	0.7	0.7	1.6	2.1	-
4	I	3.3	3.0	1.0	0.3	0.3	-
	II	8.7	7.3	2.1	0.5	0.7	-
	III	0.2	0	0.6	1.3	3.5	-
8A	I	7.0	3.7	2.3	0.7	-	-
	II	7.0	7.4	5.3	1.8	-	-
	III	0	0.2	0.3	0.2	-	-
Commercial A	I	-	-	-	-	-	0.6
	II	-	-	-	-	-	7.1
	III	-	-	-	-	-	0
Control	I	3.7	3.7	3.0	3.7	3.7	3.7
	II	10.6	10.3	10.6	10.7	8.7	8.9
	III	0	0	0	0	0	0

Example 4

Composition numbers 9, 10, 11 and 13 as well as commercial contact type sucker control material A were applied to Delhi 76 tobacco plants at the end of July at Delhi,

liters of
ined in 16.8
Delhi 34
in July.
eight and leaf
n Table 4.

v		
6	8.0	16.8
0	0.3	-
3	0.1	-
6	2.1	-
3	0.3	-
5	0.7	-
3	3.5	-
7	-	-
3	-	-
2	-	-
	-	0.6
	-	7.1
	-	0
7	3.7	3.7
7	8.7	8.9
	0	0

s well as
\ were applied
Delhi,

Ontario. Phytotoxicity ratings were taken on August 18th and sucker counts [illegible] days later. Results have been tabulated in Table 5.

Table 5

Composition Number	Rate L/ha	Sucker Control Efficacy Test Parameters I	II	III
9	6.0	6.6	30	0.62
	9.0	4.6	18	1.12
10	6.0	7.0	34	0.37
	9.0	6.7	39	0.50
11	6.0	7.4	33	0.37
	9.0	6.2	31	0.25
13	3.0	5.9	30	2.00
Commercial A	16.8	0.2	3	0.12
Hand-suckered	-	11	48	0
LSD 0.05	-	2.3	20	0.44

Example 5

Composition number 12 and a commercial contact type sucker control material B were applied to Delhi 76 tobacco plants at Delhi, Ontario on August 6. Both materials were applied at a spray volume of 450 liters per ha. at 100 KPa, and a ground speed of 6 kph, using 3x 8008 nozzles per row for composition 12, and 2x TG-3 and 1x TG-5 nozzle for commercial material B; the latter being the more common nozzle arrangement used by commercial applicators or growers. Application was made to two replications of one 24.4 meter row

per treatment. As no hand suckered control plot was run concurrently on the same experimental plot, comparison has been made in Table 7 comparing two commercial fatty alcohol based sucker control agents with a hand suckered control for tobacco grown in an adjuvent experiment plot. The crop history for the trials reported in Tables 6 and 7 have been listed as Tables 8 and 9, respectively.

Table 6

Composition Number	Rate L/ha	Sucker Control Efficacy I	II	III	Leaf Analysis IV	V	Agronomics VI	VII	VIII
12	4.5	1.5	3.1	0.50	23.6	2.41	3.57	2306	8225
Commercial B	16.8	1.6	7.9	0.13	23.7	2.44	3.57	2185	7802

Table 7

Composition Number	Rate L/ha	Sucker Control I	II	Leaf Analysis IV	V	Agronomics VI	VII	VIII
Commercial C	19.6	2.0	14	25.2	2.27	3.50	2404	8654
Commercial B	16.8	2.2	15	23.9	2.28	3.62	2346	8488
Hand-Suckered	-	15.0	39	23.6	2.33	3.56	2152	7650

Table 8

CROP HISTORY

Ploughing date	April 27
Herbicide	April 29, Tillam @ 7.7 L/ha
Nematicide	May 10, Vorlex @ 45 L/ha (row application)
Cutworm control	May 20, Ripcord @ 247 mls/ha
Variety	Newdel
Planted	May 27
Fertilizer	3-9-18 @ 950 kg/ha plus 252 kg/ha (sidedressed June 29)
Irrigation	July 17-38 mm, July 24-38 mm, August 18-38 mm
Hornworm control	August 3, Orthene @ 770 g/ha
Topped	August 2
Sprayed	August 6 (large suckers removed before spraying)
Phytotoxicity ratings	August 16
Sucker counts	August 26
Harvests	August 13, 19, 28, 31 and September 8
Frost date	August 29

Table 9

1982 CROP HISTORY

Ploughed	April 26
Herbicide	April 28, Tillam @ 5.5 kg (active)/300 L/ha
Nematicide	May 10, Vorlex in row, @ 45 L/ha
Cutworm control	May 20, Ripcord @ 247 ml/ha
Planted	May 29
Variety	Delhi 76, seedbed seedlings
Fertilizer	3-9-18 @ 950 kg/ha at planting
	3-9-18 @ 250 kg/ha, sidedressed on June 28
Irrigation	July 16, 23, and August 19 @ 38 mm each
Hornworm control	August 13, Orthene @ 770 g/ha
Topped	July 31
Sprayed	July 30, August 6 and 11
Ridomil	May 19, 25% WP 4.48 kg/ha (soil)
Harvested	August 11, 17, 23, September 2 and 8
Sucker counts	August 26
3 tip leaf measurements	August 23
Fall Frost	August 29

One aspect of this invention is that alkoxylated alcohols or their blends having an HLB value higher than 9 are self-emulsifying or soluble in water up to at least 5% v/v level thus rendering them suitable for use as sucker control agents without the addition of added inert emulsifier. Table 10 compares the emulsion stability of a selection of such alkoxylated alcohols in comparison to a commercial contact type sucker control material. In the majority of cases, the emulsion stability from the alkoxylated alcohol is superior thereby ensuring more uniform application of the control material.

Table 10

Emulsion Stability Data

Composition Number	Dilution v/v	Water Hardness	% Separation with Time (h) 0.25	0.5	1.0	2.0
Commercial A	4%	342 ppm	1.5	2.0	3.0	5.0
12	4%	"	0	0.5	3.0	5.0
	1%	"	0	0	0.5	0.5
11	4%	"	0	0	1.0	1.0
10	4%	"	0	0.8	1.8	2.0
9	4%	"	2.0	4.0	4.0	5.0
3	4%	"	0	0	0	0

A review of the comparative performance of the alkoxylated alcohols tested suggests that the optimum HLB value for best sucker control efficacy, dispersion stability and non-phytotoxicity appears to be approximately 10. It can be seen that the alkyl chain length and stereo chemistry of the parent alcohol has an effect on both the phytotoxicity and sucker control functionality of the alkoxylated alcohol. It is contemplated that closely related alkoxylated adducts of

1-octanol within the limits of this disclosure would essentially perform as well as polyoxyethylene (3.0) 1-octanol.

The preferred alkoxylating agent would appear to be ethylene oxide. We could find no advantage to the inclusion of propylene oxide in the molecule. However, the illustrative examples shows that compounds within the range of this disclosure containing propylene oxide do act as sucker control agents and as such should not be excluded from this invention. It is contemplated that alkoxylated alcohols containing butylene oxide may also perform as tobacco sucker control agents.

Some of the alkoxylated alcohols disclosed in this invention are available as commercial products, being sold primarily for their surface active agent properties. Others are not commonly commercially available. All materials utilized in the trials reported in this text were synthesized in a similar manner to the method listed below for composition 12:

Although evaluations were conducted with Delhi 76 and 34 varieties of tobacco, the invention i, not restricted to those specific varieties. Delhi 76 and 34 are among the three recommended varieties for Ontario since these varieties together with Virginia 115 are considered superior to other varieties of flue-cured tobacco in most respects. Studies have shown that contact type desuckerants are relatively unaffected by tobacco variety and thus it is contemplated that the compositions of this invention will inhibit the growth of axillary buds on other licensed varieties of flue-cured tobacco.

ABSTRACT

The growth of suckers is inhibited in flue-cured tobacco plants by applying to the plant shortly before or after topping an aqueous dispersion of alkoxylated C_6-C_{10} alcohols having an HLB of 8-14.

052083SMS01

-cured
re or after
alcohols

CLAIMS

The claims defining the invention are as follows:

1. A method for inhibiting the growth of suckers in flue-cured tobacco plants which comprises applying to the plant shortly before or after topping an effective amount of a liquid alkoxylated saturated C6-C10 alcohol having a hydrophilic-lipophilic balance value of 8-14, wherein said alkoxylated alcohol has the general formula $RO(CH_2CHR^1O)_nH$ wherein R is an alkyl group having 6-10 carbon atoms, R^1 is H, CH_3, C_2H_5 and n is 2-10.

2. The method of Claim 1 wherein the alkoxylated alcohol is applied as an aqueous dispersion.

3. A method of Claim 2 wherein said aqueous dispersion includes an effective amount of a phytobland emulsifier or surfactant.

4. The method of Claim 1 wherein there is applied to the tobacco plant an aqueous dispersion of polyoxyethylene 1-octanol with an average oxyethylene chain length of approximately 3 moles.

5. A method according to any one of claims 1 to 4 inclusive wherein said liquid alkoxylated saturated alcohols are applied at dose rates of 0.5 to 6 liters per hectare.

6. A method according to any one of claims 2 to 5 inclusive wherein said dispersion is applied at a dose rate of about 30 mls per plant.

7. A method of Claim 1 wherein said saturated alcohol is selected from the group consisting of 1 hexanol, 1-octanol, 2-octanol, 1-decanol, and trimethyl heptanol.

8. A method of Claim 3 wherein said surfactant is polysorbate-80.

9. A method according to any one of claims 1 to 8 inclusive substantially as herein described with reference to the Examples.

DATED this 6th day of June 1985

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